JUNWHA HONG

Research Officer (First Lieutenant), Republic of Korea Air Force

qbit@unist.ac.kr · junwha.github.io · Google Scholar · GitHub

RESEARCH INTERESTS

Computer Architecture and Systems for Emerging Workloads; Heterogeneous/Reconfigurable Computing.

EDUCATION

Ulsan National Institute of Science and Technology (UNIST) Mar 2020 – Feb 2024
B.S. in Computer Science and Engineering, ***summa cum laude***

GPA: 4.13/4.3 (98.3/100), Ranked 4th out of 296 students

- *Relevant Coursework: Tensor Processor Design for Image Recognition, Building Customized Computers, Parallel Computing, Computer Architecture, Cloud Computing.*

PUBLICATIONS

[1] **How Far Can PIMs Replace GPUs in Energy-Efficient Multi-Turn Agent Serving?**
Junwha Hong, Guseul Heo, Hyunsu Ye, Jongse Park, Olivia Hsu
IEEE International Symposium on Workload Characterization (IISWC), 2026; short paper.

[2] **Understanding and Exploiting Diagonal Attention Sparsity in Autoregressive Image Generation**
Daeun Kim, **Junwha Hong**, Changhun Oh, Yoonsung Kim, Yoonhyeong Lee, Jongse Park
IEEE International Symposium on Workload Characterization (IISWC), 2026.

[3] **Tortoise and Hare Guidance: Accelerating Diffusion Model Inference with Multirate Integration**
Yunghee Lee, Byeonghyun Pak, **Junwha Hong**, Hoseong Kim
Conference on Neural Information Processing Systems (NeurIPS), 2025.

[4] **CMASan: Custom Memory Allocator-aware Address Sanitizer**
Junwha Hong, Wonil Jang, Mijung Kim, Lei Yu, Yonghwi Kwon, Yuseok Jeon
IEEE Symposium on Security and Privacy (S&P), 2025 ***(Acceptance rate: 14.3%)***.

RESEARCH EXPERIENCE

Independent Research Collaboration May 2025 – Present
CMU SysCALL · KAIST CASys

Advisors: Prof. Olivia Hsu and Prof. Jongse Park.

- Characterized six multi-turn agent workloads and identified **fresh-prefill demand and live-session KV working sets** as key constraints on GPU-to-PIM replacement. Evaluated how KV-cache eviction and Session-KV residency affect energy–latency tradeoffs.
- Extended LLMServingSim for long-context and memory-pressure studies, including an **analytical communication backend (5.3× faster, ∼2% error vs. Astra-Sim)**. Built a configurable simulator for workload–hardware design-space exploration.
- Proposed the **spatial-locality hypothesis behind diagonal attention** in autoregressive image generation and identified extreme sparsity as a key evaluation regime (2nd author, IISWC'26).

Agency for Defense Development (ADD) Mar 2024 – Present
Research Officer (First Lieutenant)

UNIST S2Lab Dec 2020 – Nov 2024
Undergraduate Researcher

Advisor: Prof. Yuseok Jeon.

- Analyzed custom memory allocators across 100+ real-world C/C++ projects, **generalized AddressSanitizer false negatives into Arena and Recycler patterns**, and defined CodeQL rules for CMA identification based on common allocator-operation patterns.
- Designed **Clang/LLVM instrumentation and an allocator-aware ASan runtime** with redzones and per-instance quarantine, preserving allocator semantics at 1.09× average runtime.
- Refined testing and root-cause analysis methods to target CMA-related execution paths, uncovering **19 previously unknown memory bugs, including 7 CVEs** (one critical PHP vulnerability).

HONORS

Top Prize (Minister of Science and ICT Award) 2022
Millitech Challenge, Republic of Korea.

National Science & Technology Scholarship 2022

Excellence Prize 2021
UNIST-POSTECH-KAIST Data Science Competition.

LEADERSHIP & TEACHING

Hacker's EXciting Academy (HeXA), UNIST Sep 2020 – Feb 2024
President (2021)
- Organized technical study groups, guest lectures, and peer-led projects during COVID-19.

Student Council, UNIST Sep 2021 – Dec 2022
Design & Information Team Lead (2022); Student Platform Development TF Lead (2021)
- Built a student-information website to make campus information accessible without social media.

Teaching Assistant / Lecturer
AI Programming Foundation/I/II (Summer/Fall 2021, Spring/Fall 2022)

Instructor
CSE ABC Camp (Winter 2023/2024), Python Campus (2021), C/Arduino Summer Camp (Summer 2020)

TECHNICAL SKILLS

Programming Languages: C/C++, Python, Rust, Scala

Compiler & Hardware Tools: LLVM IR, CuTe, Chisel (Rocket Chip), Vivado HLS, TI C7x